

EMPORIA, KS, USA

WMO: 724556

Lat: 38.329N

Lon: 96.195W

Elev: 1196

StdP: 14.07

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 13989

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.8	9.7	-4.2	4.6	8.3	0.3	5.8	12.2	28.7	46.0	26.5	42.4	9.3	330	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	98.9	74.7	94.9	74.9	91.4	74.5	78.2	90.9	77.2	89.8	76.1	87.9	13.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.8	136.2	84.7	73.3	129.5	83.3	72.5	125.8	82.3	42.6	91.2	41.4	89.7	40.3	87.9	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.5	24.7	21.7	-3.0	102.0	4.9	3.8	-6.5	104.7	-9.4	106.9	-12.1	109.0	-15.7	111.7
			-3.8	80.5	4.7	1.0	-7.1	81.2	-9.9	81.8	-12.5	82.4	-15.9	83.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	56.1	31.6	35.3	45.6	55.9	65.5	74.4	79.2	69.8	57.6	45.3	34.1
		DBStd	19.04	11.40	11.43	11.19	9.35	7.81	5.94	5.67	5.97	7.55	9.29	10.03
		HDD50	2004	575	426	217	47	2	0	0	0	35	203	499
		HDD65	4721	1036	832	607	296	93	6	1	36	264	593	957
		CDD50	4230	4	14	79	224	481	733	906	859	593	269	61
		CDD65	1473	0	0	4	23	107	289	441	395	179	34	1
		CDH74	15301	0	2	50	265	934	2843	4927	4186	1751	322	21
		CDH80	6835	0	0	6	58	260	1144	2495	2076	707	87	2
(14)	Wind	WSAvg	10.5	10.6	11.0	12.1	12.9	10.9	10.3	9.0	8.4	9.3	10.3	10.9
(15)	Precipitation	PrecAvg	35.6	0.9	1.3	2.2	3.4	5.0	4.8	4.5	3.8	3.5	2.7	2.1
(16)		PrecMax	49.4	2.8	3.2	5.0	8.4	9.4	9.6	15.1	8.0	7.6	6.9	10.0
(17)		PrecMin	20.7	0.1	0.0	0.5	0.7	0.9	1.2	0.4	0.6	1.3	0.1	0.0
(18)		PrecStd	7.4	0.7	0.9	1.3	1.9	2.2	2.4	3.3	2.0	2.0	1.8	2.1
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	65.5	71.4	79.5	85.9	89.8	97.5	102.4	102.2	97.6	87.6	77.0	66.3
(20)		MCWB	52.0	53.8	62.0	66.0	71.4	75.8	74.1	73.2	73.0	69.2	60.1	55.9
(21)		DB	59.3	64.4	73.7	81.0	85.8	92.0	99.1	98.7	91.0	81.9	71.6	60.2
(22)		MCWB	47.4	51.9	58.3	63.6	70.3	75.1	75.3	74.2	72.6	65.1	58.1	50.6
(23)		DB	54.0	58.8	69.0	76.7	82.8	89.5	95.5	94.5	87.9	77.0	65.9	54.4
(24)		MCWB	44.8	49.3	56.0	61.0	69.3	75.0	75.5	74.6	71.7	63.2	54.3	46.6
(25)		DB	48.3	53.6	64.0	72.4	79.5	86.8	91.7	90.8	83.8	72.8	61.2	49.9
(26)		MCWB	41.0	45.4	53.7	60.0	67.6	73.5	75.6	73.9	69.8	61.9	52.6	43.5
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	56.1	58.8	65.6	68.9	75.3	79.2	79.8	79.3	76.7	72.5	63.8	59.5
(28)		MCDB	60.6	65.0	76.2	81.0	85.1	91.3	92.9	90.5	88.2	81.8	71.0	64.0
(29)		WB	49.6	54.2	62.0	66.4	73.2	77.5	78.4	77.7	74.8	69.6	59.7	52.7
(30)		MCDB	57.9	61.3	68.6	75.8	82.6	88.9	91.6	90.3	86.2	77.0	66.8	57.0
(31)		WB	45.2	50.0	58.9	64.2	71.3	76.1	77.4	76.4	73.3	66.8	57.0	47.4
(32)		MCDB	52.4	58.7	66.4	72.8	79.8	86.7	90.8	89.0	84.3	73.1	63.8	53.3
(33)		WB	41.3	45.0	54.7	61.6	69.2	74.7	76.4	75.1	71.5	63.7	53.6	43.6
(34)		MCDB	48.8	52.9	63.0	70.8	77.3	84.5	88.7	87.0	81.1	70.6	60.1	49.4
(35)	Mean Daily Temperature Range	MDBR	20.0	20.5	22.3	22.2	20.2	19.9	21.0	21.7	22.4	22.3	21.5	19.3
(36)		5% DB	28.2	29.9	29.7	28.3	23.1	22.1	25.6	26.5	25.4	26.9	28.3	26.7
(37)		MCWBR	18.0	19.0	17.0	15.3	11.7	9.6	8.0	8.4	10.0	13.4	16.3	18.0
(38)		5% WB	26.9	27.6	23.8	23.9	20.3	20.6	21.5	22.4	22.0	21.4	23.5	24.2
(39)	Clear-Sky Solar Irradiance	taub	0.286	0.300	0.333	0.377	0.414	0.426	0.429	0.430	0.382	0.339	0.307	0.284
(40)		taud	2.488	2.461	2.414	2.320	2.280	2.269	2.303	2.282	2.375	2.480	2.500	2.509
(41)		Ebn at Noon	286	296	295	286	276	271	270	266	274	278	276	279
(42)		Edh at Noon	25	29	34	40	42	43	41	41	35	29	24	23
(44)	All-Sky Solar Radiation	RadAvg	759	992	1289	1636	1850	2110	2079	1862	1543	1136	821	649
(45)		RadStd	73	128	94	105	158	144	141	102	108	128	75	59

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	-1.62	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page